Title:

Terraform

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1. What is Infrastructure as Code.

Practice which enables developers and operations teams to provision and manage infrastructure through code & instruction rather than manually configuring the process.

a) Automation: ✓

Iac allows to perform infrastructure tasks automatically via code.

- No manual interventions
- Less chances of error
- Rapid
- Streamlines repeatitive tasks.

Eg. Using Terraform to create EC2 instead of using Aws.

b) Declarative & Imperative ✓

- → tell tool what to do and desired result

tool - configures and determines the necessary steps.

tell tool how to do it by specifying the commands ① or steps ② to achieve the result.

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c) Version Control: ✓

Tracking changes made to code over time.

Supports rollback to older builds incase inconsistencies in code resulting in problem for newer builds.

d) Idempotence: ✓

Output of fixed ① configuration of code would produce exact same intended result .... ② over and over again.

Eg. Tfm code configured to create _x86 EC2 wont create Arm64 EC2

No manual config mistake.

e) Consistency: IaC creates identical environment with consistent configurations.

Reproducibility: recreation of envirn is very easy using same code.

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(2) Benefits of IaC:

- ✓ Speed and Efficiency without human intervention.
- ✓ Cost-effective: by automation.
- ✓ (**) Disaster Recovery: Recreation of environment is rapid in events of loss.
- ✓ Scalability: IaC can scale infra-automatically

(3) Practice: Terraform Code to provision EC2.

- Configure Aws IAM Role (AwsEC2full access)
- Attach to host EC2.... Install Terraform
- Mkdir terraform_cd.... write main.tf
- terraform init, plan, apply.

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4. Challenges:

→ Complexity in large-scale deployment: Managing large and complex infrastructure with IaC

→ (State Management:)

→ Security: Storing sensitive data exposed in codebase can be a security risk. Solution: using Vault.

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II. Terraform.

1. What is terraform?

Terraform is an open source ① Infrastructure as Code (IaC) tool created by HashiCorp. ②

It allows us to ① define, ② provision, and ③ manage cloud resources using a declarative config language. ✓

How terraform work:

- **Write**: (terraform init); configuration file (.tf) and initialize working directory to download the necessary provider plugins. ✓ (creates environment)
- **Plan**: (terraform plan); previews changes it will make (create, update, destroy) without actually touching infrastructure
- **Apply**: (terraform apply); Terraform executes the planed changes, provision or updating resources in cloud env.

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Terraform Vs Ansible:

Terraform infrastructure code tool — used to provision infrastructure via IaC

Ansible is infrastructure configuration tool used to configure and customised infra - after being created.

Aws (Cloud formation) = terraform

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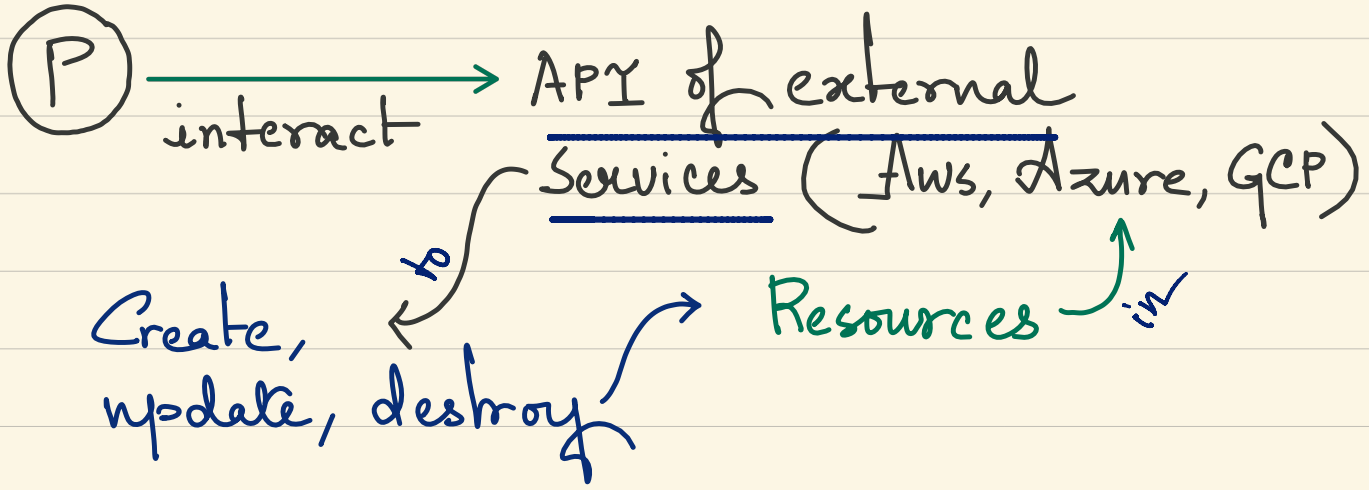
2. Provider:



Syntax:

```
provider "aws" {
    region : "us-east-1"
    secret_key : "....."
    access_key : "....."
                    }
```

3. Resources: A resource in terraform represents a specific components of our infrastructure

Its defined by: resource block.

```
resource "aws_instance" "name" {
block    parameter      nm
    ami = "ami-0511...."   arg
    instance_type = "t3.micro"
                    }
```

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4. Attributes: Resource typically include multiple attributes such as:

Argument (human config)

a. ami 'holds the ami Id'
b. instance_type 'holds the type power resource would get'
c. tags 'Special key: value pair attached to resource'

(Input ⟹ Provider API ⟹ Set up Infrastructure

Arg → Data before creating
Computed → Data after creating (Cloud provider API Config)

Computed

provide metadata and details about the resource that are obtained after resource actually exists.

Eg. value = aws_instance.web.public_ip

5. State
- what is Terraform State?

Terraform state file (terraform.tfstate) used to keep track of resources that Terraform manages. Stores metadata about your infrastructure in statefile.

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Crucial ensuring terraform understand the current state of infrastructure.

Important :-

a) State file ensures terraform (update, created or destroy) decisions code maps configuration real infrastructure

b) State Management :

By default state is locally stored in <terraform.tfstate>
by for larger environments use :

S3 for Aws.

c) Remote State :

Storing statefile remotely is beneficial for collaboration.

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Terraform State In details:

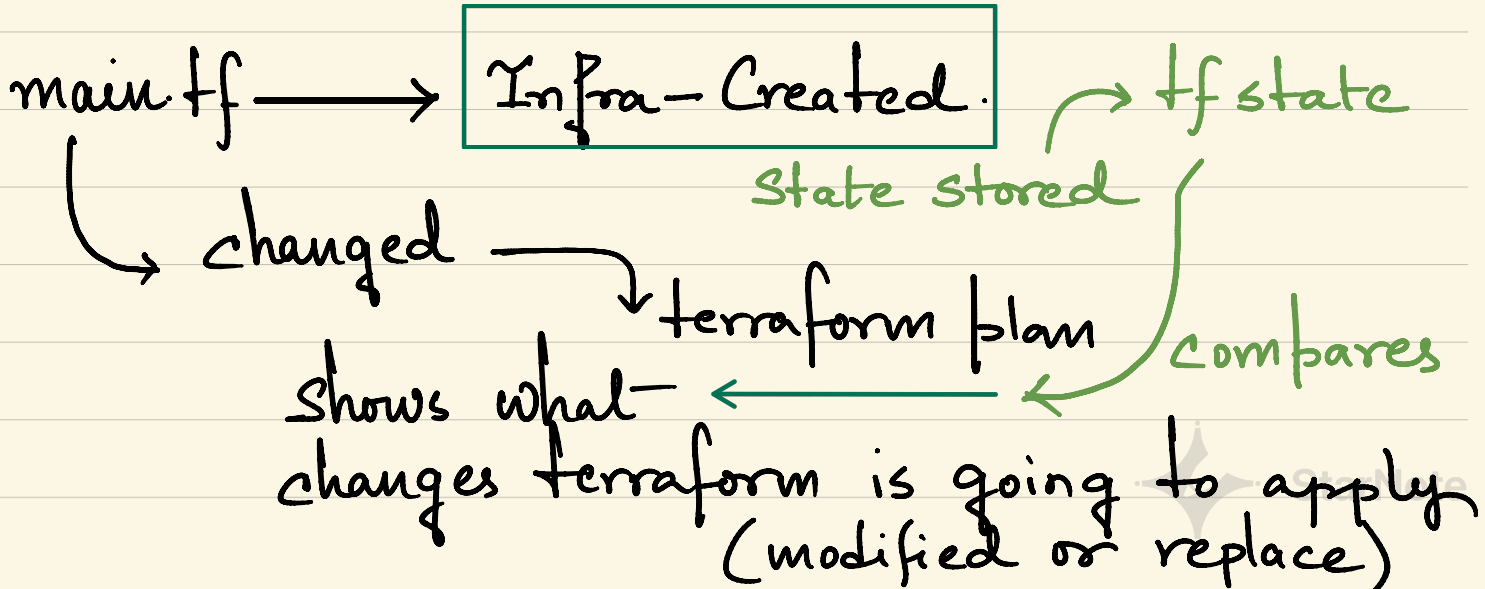
main.tf (Terraform Config)
store what we want to do with our infrastructure

terraform.tfstate (terraform State)
store what terraform knowns about our infrastructure.

Aws infrastructure
(resources running)

Use of State file:
- Resource tracking
- Change detection
- Collaborations.

Example on how stateworks.



State locking

State locking: is a mechanism that prevents multiple-users modify the same state simultaneously (No concurrent changes)

Problem: operations may overlap, conflicting changes, mismatched configuration, leading to state inconsistency

How terraform uses statelocking

- dev A: starts Apply → Terraform acquires lock → shared state
- dev B: tries to Apply → operation awaits completion (A) or runs into lock-related error

lock is released once dev A's task is completed.

backend.tf

```
terraform {
  backend "s3" {
    bucket = "..."
    key    = "..."
    region = "..." }
}
```

Commands:

use_lockfile = true

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Scenarios :

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6. Terraform Configuration files: (HCL)

- ✓ Provider Block: (Specifies cloud provider)

```
provider "aws" {
  region = " " }
```

- Resource Block: (Defines what resource to manage)

```
resource "aws_instance" "name" {
  ami = "111
  instance type = " "
}
```

- ✓ Variable block: allows → declare configs; make configurations dynamic and reusable

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```
variable "instance type" {
    default = "t3.micro"
}
```

- Output block:

Output values to console after running terraform apply

```
output "instance_ip" {
    value = aws_instance.example.
            public_ip.
```

- Data Sources:
  Allows us to retrieve information about existing infrastructure can also be used in configuration.

7. Terraform Workflow:

- init: initializes working directory (→ Sets up environment) downloads necessary provider plugins configures backend if using remote state.

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- validate: • Syntax & Internal Consistency (invalid argument/ incorrect d-type) checking
  - Attribute & variable checks.
  - Isolated verification checks performed exclusively at local/offline
- plan: Show preview of actions terraform is about to perform and output
  (Create(Addition), modify(changes), remove(deletion).
- apply: Applies actions specified by the plan to the infrastructure
- destroy: Destroys infrastructure maintained by terraform. [Cleanup]

8. Variables:

variables in terraform provides versatility and reusability while configuring.

Syntax: variable "name" {
description = ""
default = "" }

How to use: <var.name>

Ways to override variables:
> commandline: -var="region=val"
> terraform.tfvars file.

9. Modules:

Collection of Terraform configuration files grouped together to create and manage AWS resources

Examples:— 5 EC2 instances with same configuration
→ Create 1 EC2, reuse it 5 times

- Reusability
- Organization
- Consistency
- Easy Maintenance:

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Structure of Module:

folder: (local)
- ↳ main.tf
- ↳ variables.tf
- ↳ Outputs.tf

(Remote terraform registry)

```
module "vpc" {
source = "./module/
          vpc"
cidr_block = " "
}
```

1. **Root Module:** Main directory where init, apply are called. It calls other modules when required

2. **Child Module:** Module which is called by the root, contains resuable configs EC2, S3, or VPC.

3. **Published or external module:** resuable module obtained from terraform registry.

(Note:)

Avoid storing sensitive information directly in terraform files.

Good practice to use services like Aws Secret Manager, HasiCorp Vault or terraform cloud to securely manage secrets.

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10. Provisioners & Lifecycle:

Provisioners: allows you to execute scripts or commands on resource after they have been created

- Types of provisioners:
  - remote-exec: Executes commands over SSH
  - local-exec: Executes commands locally

lifecycle:

- prevent_destroy: Prevent accidental destruction of resource.
- ignore_changes: Ignore changes to certain attributes of resource.

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11. Terraform - Interpolation:

Interpolation in Terraform → inserting the value of a [variable, resource attribute, or expression into a string or configuration

Helps to use dynamic values instead of using fixed / hardcoded values.

old Syntax: ${var.instance_name}

New Syntax: var.instance_name.

Example

```
resource "aws-default-vpc" default {
  id:
}

resource "aws-security-group" my-sg {
  vpc_id = aws_default-vpc.default.id
```

..... dynamic value binding.

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